An Effective Orthogonal-Bimetric Negentropic-Sector Model for Dark Matter and Cosmic Acceleration

Pascal Ranaora[1,*]

[1] *Information Physics Institute - Independent Researcher, Sydney, Australia*

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We propose an effective Orthogonal-bimetric framework in which dark matter and cosmic acceleration arise as two distinct real gravitational projections of an orthogonal informational sector. The construction is based on an orthogonal bimetric tensor space $\mathcal{G}_{MN} = \mathrm{diag}(g_{\mu\nu}, f_{\mu\nu})$, where $g_{\mu\nu}$ denotes the ordinary visible-sector spacetime metric and $f_{\mu\nu}$ denotes an independent negentropic metric associated with maintained informational order. This block-diagonal matrix structure formally enforces the mathematical orthogonality between thermodynamic embodiment and informational geometry, avoiding trivial Orthogonal superpositions. The Standard Model is coupled minimally to $g_{\mu\nu}$, while a scalar-order sector and massive informational excitations propagate on $f_{\mu\nu}$. Localized excitations behave as pressureless dark matter, whereas the vacuum energy of the order parameter behaves as an effective negative-pressure component capable of sourcing cosmic acceleration. The effective graviton mass bridging the two metrics is dynamically generated via a Hassan–Rosen-type bimetric interaction, modulated by the negentropic scalar. We further introduce an emergy-inspired interpretation of the real sector and a Orthogonal value functional combining thermodynamic memory and informational order. The resulting model is presented as a phenomenological effective field theory, with consistency conditions from ghost freedom, stability, gravitational-wave propagation, and cosmological observables.

I. INTRODUCTION

The standard cosmological model, ΛCDM, successfully describes a wide range of observations using two dominant dark components: cold dark matter and dark energy. Dark matter behaves as a pressureless clustering component, while dark energy is consistent with a nearly homogeneous component with equation of state $w \simeq -1$. Despite this phenomenological success, the microscopic origin of both components remains unknown.

A number of theoretical developments suggest that spacetime geometry, entropy, and information may be closely related. These include black-hole thermodynamics, holographic arguments, and emergent-gravity approaches. However, these frameworks do not by themselves provide a minimal phenomenological model capable of reproducing the observed dark sector while remaining testable.

In this work, we introduce an effective negentropic-sector model based on a bimetric structure, in which the physical metric $g_{\mu\nu}$ is supplemented by an additional metric $f_{\mu\nu}$. Because physical mass-energy and negentropic information obey fundamentally different conservation laws, they cannot be simply summed in the Orthogonal plane. We formally group these into an orthogonal bimetric state tensor,

$$\mathcal{G}_{MN} = \begin{pmatrix} g_{\mu\nu} & 0 \\ 0 & f_{\mu\nu} \end{pmatrix}, \tag{1}$$

which serves as the fundamental representation of the two coupled geometric sectors. While they share the same 4D spacetime manifold to allow for Hassan-Rosen mixing, the block-diagonal structure ensures they operate on independent tangent subspaces. Physical observables remain embedded within the projection of $g_{\mu\nu}$, while $f_{\mu\nu}$ encodes the orthogonal informational sector.

The negentropic sector contains two dominant degrees of freedom: a massive excitation X, which behaves as cold dark matter at the background level, and a slowly evolving scalar field χ, whose potential energy can generate accelerated expansion. The model is therefore a minimal extension of ghost-free bimetric constructions, augmented by an interpretive framework in which the $f_{\mu\nu}$ manifold is associated with maintained informational order.

To make this interpretation operational, we frame the orthogonal informational sector as a macroscopic dissipative structure. Drawing on non-equilibrium thermodynamics, we posit that irreversible physical work in the visible sector stabilizes persistent informational states in the orthogonal sector. The maintenance of this order provides a geometric mechanism for the effective mass of the bimetric graviton.

The goal of this paper is twofold: first, to define a minimal effective bimetric model capable of reproducing the ΛCDM dark sector dynamically; and second, to formulate a generalized thermodynamic framework where the persistent order of the informational metric is sustained by the continuous dissipation of physical exergy.

* pascal.ranaora@informationphysicsinstitute.net

II. ORTHOGONAL-BIMETRIC GEOMETRIC STRUCTURE

We introduce two Lorentzian metrics defined over the same 4D coordinate manifold:

$$g_{\mu\nu} \qquad \text{and} \qquad f_{\mu\nu}. \tag{2}$$

The metric $g_{\mu\nu}$ is the physical metric seen by Standard Model matter. The metric $f_{\mu\nu}$ is the negentropic-sector metric.

The global state manifold is defined formally via the direct sum of the two metrics into an 8D block-diagonal state tensor:

$$\mathcal{G}_{MN} = g_{\mu\nu} \oplus f_{\mu\nu} = \begin{pmatrix} g_{\mu\nu} & 0 \\ 0 & f_{\mu\nu} \end{pmatrix}. \tag{3}$$

This matrix formulation guarantees that no off-diagonal mixing occurs kinematically; the metrics interact exclusively through the Hassan–Rosen dynamic bimetric potential rather than linear superposition. Physical observables are obtained by projecting the state tensor onto the physical subspace,

$$\mathcal{O}_{\text{phys}} = \text{Proj}_g \left[\mathcal{O}\left(\mathcal{G}_{MN}\right)\right]. \tag{4}$$

The ordinary spacetime interval is

$$ds_g^2 = g_{\mu\nu} dx^\mu dx^\nu, \tag{5}$$

while the informational-sector interval is

$$ds_f^2 = f_{\mu\nu} dx^\mu dx^\nu. \tag{6}$$

No Standard Model particle is assumed to follow geodesics of $f_{\mu\nu}$. The orthogonal informational sector couples to visible matter only gravitationally or through small portal interactions.

III. ACTION

The full effective action is written as

$$S = S_g + S_f + S_{\text{mix}} + S_{\text{SM}} + S_{\mathcal{I}} + S_{\text{portal}}. \tag{7}$$

The visible gravitational action is

$$S_g = \frac{M_g^2}{2} \int d^4x \sqrt{-g} R[g]. \tag{8}$$

The informational gravitational action is

$$S_f = \frac{M_f^2}{2} \int d^4x \sqrt{-f} R[f]. \tag{9}$$

The Standard Model action is minimally coupled to $g_{\mu\nu}$,

$$S_{\text{SM}} = \int d^4x \sqrt{-g} \mathcal{L}_{\text{SM}}\left(\psi_{\text{SM}}, g_{\mu\nu}\right). \tag{10}$$

The orthogonal information-sector action is

$$S_{\mathcal{I}} = \int d^4x \sqrt{-f} \left[-\frac{1}{2} f^{\mu\nu} \nabla_\mu^{(f)} \chi \nabla_\nu^{(f)} \chi - U(\chi) + \mathcal{L}_X \right], \tag{11}$$

where χ is a slowly varying negentropic order parameter and X denotes massive informational excitations.

A minimal example for the massive field is

$$\mathcal{L}_X = -\frac{1}{2} f^{\mu\nu} \nabla_\mu^{(f)} X \nabla_\nu^{(f)} X - \frac{1}{2} m_X^2 X^2 - \frac{\lambda_X}{4} X^4. \tag{12}$$

The bimetric mixing action is chosen to be of the Hassan–Rosen type to avoid the Boulware–Deser ghost at the effective level. Crucially, we promote the bimetric interaction scale to a dynamical function of the order parameter χ, such that the coupling between the two metrics is mediated by the density of maintained informational order:

$$S_{\text{mix}} = m^2(\chi) M_{\text{eff}}^2 \int d^4x \sqrt{-g} \sum_{n=0}^{4} \beta_n e_n \left(\sqrt{g^{-1} f}\right), \tag{13}$$

where e_n are the elementary symmetric polynomials of the matrix square root $\sqrt{g^{-1} f}$. In this framework, the effective mass of the massive spin-2 excitation is dynamically governed by the negentropic order parameter χ. We parameterize this coupling via a conformal-like exponential factor, commonly employed in scalar-tensor theories:

$$m^2(\chi) = m_0^2 \exp\left(-\frac{\beta \chi}{M_{\text{Pl}}}\right), \tag{14}$$

where m_0 is the bare graviton mass and β is a dimensionless coupling constant. Through this relation, the escalation of informational order dynamically modulates the interaction strength between the physical and negentropic metrics, physically manifesting the gravitational consequence of informational inertia.

The portal action is assumed to be small,

$$S_{\text{portal}} = \int d^4x \sqrt{-g} \left[-\frac{\lambda_{H\chi}}{2} H^\dagger H \chi^2 - \frac{\lambda_{HX}}{2} H^\dagger H X^2 + \cdots \right]. \tag{15}$$

The smallness of $\lambda_{H\chi}$ and λ_{HX} explains why the orthogonal informational sector is not directly observed in laboratory physics.

IV. FIELD EQUATIONS

Variation of the action with respect to $g_{\mu\nu}$ yields the field equations for the physical sector, where $G_{\mu\nu}[g]$ is the standard Einstein tensor constructed from the g-metric:

$$M_g^2 G_{\mu\nu}[g] = T^{\rm SM}_{\mu\nu} + T^{\rm portal}_{\mu\nu} + T^{{\rm mix},g}_{\mu\nu}. \tag{16}$$

Similarly, variation with respect to the negentropic metric $f_{\mu\nu}$ yields the informational field equations, with $G_{\mu\nu}[f]$ denoting the Einstein tensor constructed from the f-metric:

$$M_f^2 G_{\mu\nu}[f] = T^{\mathcal{I}}_{\mu\nu} + T^{{\rm mix},f}_{\mu\nu}. \tag{17}$$

The informational-sector stress-energy tensor is

$$T^{\mathcal{I}}_{\mu\nu} = \nabla^{(f)}_\mu \chi \nabla^{(f)}_\nu \chi - f_{\mu\nu}\left[\frac{1}{2} f^{\alpha\beta}\nabla^{(f)}_\alpha \chi \nabla^{(f)}_\beta \chi + U(\chi)\right] + T^X_{\mu\nu}. \tag{18}$$

Because the bimetric mixing scale $m^2(\chi)$ is dynamically coupled to the order parameter, the variation of $S_{\rm mix}$ with respect to χ yields a critical effective source term. The scalar order field obeys:

$$\Box_f \chi - \frac{dU}{d\chi} + \frac{1}{\sqrt{-f}}\frac{\delta S_{\rm mix}}{\delta \chi} = J_{\rm portal}, \tag{19}$$

where $J_{\rm portal}$ is a small source induced by portal couplings. The mixing derivative acts as a geometric restorative force, explicitly demonstrating that the cosmological interaction between the visible and negentropic metrics actively drives the dynamics of the informational vacuum, thereby stabilizing the cosmic acceleration.

The informational excitation X obeys

$$\Box_f X - m_X^2 X - \lambda_X X^3 = J_X. \tag{20}$$

In the limit of negligible portal interactions, $J_{\rm portal} \approx 0$ and $J_X \approx 0$.

V. COSMOLOGICAL BACKGROUND

We consider homogeneous and isotropic backgrounds,

$$ds_g^2 = -dt^2 + a^2(t) d\vec{x}^{\,2}, \tag{21}$$

$$ds_f^2 = -N_f^2(t) dt^2 + b^2(t) d\vec{x}^{\,2}. \tag{22}$$

Crucially, the lapse function $N_f(t)$ governs the rate at which informational time flows relative to physical Newtonian time t. In our framework, this is exactly the topological block rate $d\theta/dt$. Therefore, $N_f(t)$ is not a free gauge parameter, but the geometric manifestation of the Difficulty Adjustment Algorithm (DAA) enforcing chronological homeostasis.

The visible Hubble rate is

$$H_g = \frac{\dot a}{a}. \tag{23}$$

The informational-sector Hubble rate is

$$H_f = \frac{1}{N_f}\frac{\dot b}{b}. \tag{24}$$

The visible Friedmann equation takes the schematic form

$$3M_g^2 H_g^2 = \rho_b + \rho_r + \rho^{\rm eff}_{\rm DM} + \rho^{\rm eff}_{\rm DE} + \rho_{\rm mix}. \tag{25}$$

The effective dark matter density is associated with massive informational excitations,

$$\rho^{\rm eff}_{\rm DM} \equiv \rho^{\rm proj}_X. \tag{26}$$

For nonrelativistic excitations,

$$p_X \ll \rho_X, \tag{27}$$

and therefore

$$\rho_X \propto b^{-3}. \tag{28}$$

Because the Hassan–Rosen bimetric interaction forces the two metrics to expand proportionally at the background level to avoid ghost instabilities (maintaining a stable ratio $b(t) \propto a(t)$), the dilution in the informational space mirrors the physical expansion. When projected into the visible sector, this behaves approximately as cold dark matter,

$$\rho^{\rm eff}_{\rm DM} \propto a^{-3}. \tag{29}$$

The effective dark energy density is associated with the negentropic order field,

$$\rho^{\rm eff}_{\rm DE} \equiv \rho^{\rm proj}_\chi. \tag{30}$$

As demonstrated in the previous section, if the physical exergy dissipation Φ_g is sufficiently large, the induced topological friction $\gamma(\Phi_g)$ brutally damps the kinetic energy of the order field. This continuous dissipation forces the order parameter into a quasi-frozen, metastable state,

$$\dot\chi^2 \ll U(\chi). \tag{31}$$

In this strongly friction-dominated regime, one obtains

$$\rho_\chi \approx U(\chi), \tag{32}$$

$$p_\chi \approx -U(\chi), \tag{33}$$

and hence

$$w_\chi = \frac{p_\chi}{\rho_\chi} \approx -1. \tag{34}$$

Thus the same orthogonal informational sector can generate both dark-matter-like and dark-energy-like behavior, provided its localized excitations and vacuum/order components are dynamically distinct.

VI. DARK MATTER AS INFORMATIONAL INERTIA

In this model, dark matter is not interpreted as ordinary invisible matter, but as the visible gravitational projection of localized negentropic excitations.

The field X carries an energy density defined on the informational manifold. Recalling that the lapse function governs the topological clock rate ($N_f = d\theta/dt$), the kinetic term strictly measures the variation of the field with respect to the discrete topological time θ, such that $\dot{X}/N_f \equiv dX/d\theta = X'$:

$$\rho_X = \frac{1}{2}(X')^2 + \frac{1}{2}m_X^2 X^2 + \frac{\lambda_X}{4}X^4. \tag{35}$$

When X oscillates coherently in a quadratic potential, its averaged pressure vanishes,

$$\langle p_X \rangle \approx 0, \tag{36}$$

and therefore its energy density dilutes with the informational spatial volume,

$$\langle \rho_X \rangle \propto b^{-3}. \tag{37}$$

Because the Hassan–Rosen interaction and the cybernetic governor (DAA) enforce a macroscopic proportional scaling between the physical and informational spatial volumes ($b(t) \propto a(t)$), the volumetric dilution in the negentropic sector directly mirrors the physical expansion. When projected into the visible sector, this behaves effectively as cold dark matter:

$$\rho_{\rm DM}^{\rm eff} \propto a^{-3}. \tag{38}$$

The visible sector feels this component exclusively through the bimetric interaction. We define the effective observable density as:

$$\rho_{\rm DM}^{\rm eff} = \Gamma_X(a, b, \beta_n)\rho_X, \tag{39}$$

where Γ_X is a projection factor dictated by the Hassan–Rosen coupling polynomials $\beta_n e_n(\sqrt{g^{-1}f})$.

Physically, X may be interpreted as localized informational inertia—macroscopic topological knots or heavily secured data structures (analogous to deeply confirmed state transitions in a ledger). Because the $f_{\mu\nu}$ manifold operates in a regime of driven negative entropy, these localized structures strongly resist deformation or erasure. This resistance to entropic decay manifests through the Hassan–Rosen mixing term as an effective inertial mass.

Because the state tensor $\mathcal{G}_{MN}$ enforces strict orthogonality, X does not couple to the Standard Model gauge groups (electromagnetism). It interacts solely via the bimetric mass matrix, contributing a restorative geometric pressure that is phenomenologically observed as Cold Dark Matter in the visible universe.

VII. COSMIC ACCELERATION FROM INFORMATIONAL ORDER

The scalar order parameter χ represents the macroscopic vacuum state of the negentropic manifold. Its energy density and pressure are derived with respect to the topological time θ. Recalling that $\dot{\chi}/N_f \equiv d\chi/d\theta = \chi'$, we have:

$$\rho_\chi = \frac{1}{2}(\chi')^2 + U(\chi), \tag{40}$$

$$p_\chi = \frac{1}{2}(\chi')^2 - U(\chi). \tag{41}$$

The equation-of-state parameter is therefore:

$$w_\chi = \frac{\frac{1}{2}(\chi')^2 - U(\chi)}{\frac{1}{2}(\chi')^2 + U(\chi)}. \tag{42}$$

In standard quintessence models, the slow-roll condition $\dot{\chi}^2 \ll N_f^2 U(\chi)$ (or $(\chi')^2 \ll U(\chi)$) is often imposed *ad hoc* to generate late-time acceleration. In our framework, this quasi-stable order phase is a rigorous dynamical consequence of the system's open thermodynamics.

As established previously, the immense physical exergy dissipation (Φ_g) generates a macroscopic geometric friction $\gamma(\Phi_g)$ through the DAA-induced covariant derivative. This thermodynamic friction severely overdamps the topological evolution of the order parameter ($\chi' \to 0$), freezing the field high on its potential well:

$$(\chi')^2 \ll U(\chi), \tag{43}$$

yielding the dark-energy equation of state:

$$w_\chi \approx -1. \tag{44}$$

The visible effective dark-energy density is perceived through the gravitational projection:

$$\rho_{\rm DE}^{\rm eff} = \Gamma_\chi(a, b, \beta_n)\rho_\chi + \rho_{\rm mix}^{\Lambda}(\chi). \tag{45}$$

Crucially, $\rho_{\rm mix}^{\Lambda}(\chi)$ denotes the vacuum contribution of the Hassan–Rosen bimetric potential. Unlike a static cosmological constant, this term is dynamically modulated by the informational order scalar via the mass coupling $m^2(\chi)$.

In the interpretive language of this paper, cosmic acceleration is not sourced by a featureless void, but by the relentless topological pressure of maintained informational order. The universe accelerates because immense physical work is being continually dissipated to prevent the negentropic geometry from collapsing under its own entropic weight.

VIII. EMERGY, INFORMATION ORTHOGONALITY, AND BIMETRIC VALUE

The diagonal tensor formulation used above admits a philosophical interpretation closely related to a two-dimensional state vector in an orthogonal phase space:

$$\vec{V} = \begin{pmatrix} a \\ b \end{pmatrix}. \tag{46}$$

The a component, corresponding to the $g_{\mu\nu}$ physical axis, represents embodied thermodynamic reality: matter, radiation, biological organization, geochemical structure, and the accumulated memory of solar work. The b component, corresponding to the $f_{\mu\nu}$ axis, represents orthogonal informational order.

This section does not introduce additional fundamental dynamics. Instead, it provides an interpretive layer connecting the effective field theory to a broader thermodynamic-information picture.

A. Emergy and the Physical Axis

We introduce a coarse-grained emergy density $\mathcal{E}_\odot$, representing history-dependent solar work stored in physical and biological structures. A phenomenological definition is

$$\mathcal{E}_\odot(x,t) = \int_{-\infty}^{t} K(t,t')\Phi_\odot(x,t')dt', \tag{47}$$

where $\Phi_\odot$ is absorbed solar-energy flux and $K(t,t')$ is a memory kernel encoding persistence, transformation efficiency, and ecological or geological storage.

The physical-axis value density may then be represented schematically as

$$a(x,t) \sim \mathcal{E}_\odot(x,t). \tag{48}$$

This identification is not meant to replace stress-energy in general relativity. Rather, it provides a thermodynamic interpretation of the physical sector as a reservoir of embodied work.

At the effective level, one may write the macroscopic stress-energy tensor as:

$$T_{\mu\nu}^{\rm phys} = T_{\mu\nu}^{\rm SM} + T_{\mu\nu}^{\rm bio} + T_{\mu\nu}^{\rm emergy}, \tag{49}$$

where $T_{\mu\nu}^{\rm bio}$ and $T_{\mu\nu}^{\rm emergy}$ are coarse-grained descriptions valid only at macroscopic ecological or planetary scales.

B. Information Orthogonality and the Independent Axis

The metric $f_{\mu\nu}$ is interpreted as the geometric carrier of orthogonal informational order. The block-diagonal structure of the state tensor,

$$\mathcal{G}_{MN} = g_{\mu\nu} \oplus f_{\mu\nu}, \tag{50}$$

denotes a $\pi/2$-like rotation in an abstract value phase space. It ensures that informational order and physical mass-energy remain mathematically distinct, interacting dynamically but never kinematically.

Recalling our topological time derivative ($\chi' = d\chi/d\theta$), we define a coarse-grained information-order density,

$$\mathcal{I} = \frac{1}{V_f}\int_{\Sigma_f} d^3x\sqrt{\gamma_f}\left[\frac{1}{2}(\chi')^2 + U(\chi)\right], \tag{51}$$

where V_f is the informational-sector spatial volume,

$$V_f = \int_{\Sigma_f} d^3x\sqrt{\gamma_f}. \tag{52}$$

A phenomenological projection from real thermodynamic work into orthogonal information order may be written as

$$\mathcal{I} = \kappa_I \Pi_\perp\left[\mathcal{E}_\odot\right], \tag{53}$$

where $\Pi_\perp$ is an abstract orthogonalization map and κ_I is an emergy-information conversion coefficient.

This expression may be understood as a formal analogue of irreversible physical work being encoded into durable information. Cryptographic proof systems, including proof-of-work architectures, provide a terrestrial analogy: they transform dissipated physical work into an ordered, difficult-to-alter informational record. In the present cosmological model, however, such systems are not assumed to be the source of dark matter or dark energy. They serve only as an illustrative analogue of the more general principle of thermodynamic work projected into maintained information order.

C. Negative Entropy as the Source of Hidden Curvature

The maintenance of macroscopic order requires the continuous export of entropy, establishing the orthogonal informational sector as an open dissipative structure. Following the Brillouin-Schrödinger paradigm, information acts as negative entropy (negentropy). The global thermodynamic constraint across the bimetric manifold requires:

$$dS_{\text{tot}} = dS_g + dS_f \geq 0, \tag{54}$$

where S_g and S_f are the entropies associated with the physical and negentropic metrics, respectively.

For the informational sector to progressively crystallize state updates, its local entropy evolution must be strictly negative:

$$\dot{S}_f < 0. \tag{55}$$

This violation of local entropy maximization is dynamically permitted only because it is coupled to massive, compensating thermal dissipation in the visible metric:

$$\dot{S}_g \gg \left|\dot{S}_f\right|. \tag{56}$$

This fundamental asymmetry has profound geometric consequences. In standard relativity, positive entropy production is associated with the dispersion of mass-energy. Conversely, the accumulation of negative entropy ($\dot{S}_f < 0$) within the informational metric implies an escalating local ordering.

We formally associate the scalar order field χ with the negentropic density of the orthogonal manifold:

$$\chi \propto \exp\left(-\frac{S_f}{k_B}\right). \tag{57}$$

Consequently, the negentropic sector does not violate the second law of thermodynamics; rather, it inverts its dynamical consequence. The negative entropy of the $f_{\mu\nu}$ metric acts as the fundamental source for the stress-energy tensor $T^{\mathcal{I}}_{\mu\nu}$, exerting a topological pressure that projects onto the visible universe as dark energy.

D. Orthogonal Value Functional

The philosophical norm associated with the orthogonal state vector

$$\vec{V} = \begin{pmatrix} a \\ b \end{pmatrix} \tag{58}$$

is its Euclidean magnitude:

$$|\vec{V}| = \sqrt{a^2 + b^2}. \tag{59}$$

In the present framework, the analogous real observable is an orthogonal value functional combining thermodynamic embodiment and informational order:

$$\mathcal{V}_{\text{HB}} = \int_{\Sigma_t} d^3x \sqrt{\gamma_g} \left[\mathcal{E}_\odot^2 + \alpha_I \mathcal{I}^2\right]^{1/2}. \tag{60}$$

Here α_I converts information-order density into the same effective units as emergy density.

The functional $\mathcal{V}_{\text{HB}}$ is not a new fundamental interaction. It is a macroscopic diagnostic of joint physical-informational value. It formalizes the idea that a stable civilization, biosphere, or complex system cannot maximize informational abstraction while destroying its real thermodynamic base.

One may express a sustainability condition as

$$\frac{d\mathcal{V}_{\text{HB}}}{dt} > 0, \qquad \frac{d}{dt}\int_{\Sigma_t} d^3x \sqrt{\gamma_g}\mathcal{E}_\odot \geq 0. \tag{61}$$

The first condition states that total bimetric value grows. The second states that such growth must not be achieved by depleting the physical-axis emergy base.

This provides a formal place for the philosophical concept of *Homo Biodiversitas*: an observer or civilization that increases informational order while preserving or regenerating the thermodynamic and ecological substrate from which that order is derived.

IX. THERMODYNAMIC DISSIPATION AND IRREVERSIBLE STATE TRANSITIONS

The orthogonal informational sector introduced above acts as an effective geometric description of maintained macroscopic order. To ground this concept phenomenologically, it is instructive to examine systems where irreversible physical work is systematically converted into persistent, orthogonal informational structures. Distributed cryptographic consensus mechanisms provide a rigorous, albeit localized, thermodynamic analogue.

We stress that Proof-of-Work plays no dynamical role in the cosmological model. Its energy scale is negligible in comparison to cosmological densities, and no direct

contribution to gravitational dynamics is implied. Its role is purely illustrative: it provides a controlled example of the mapping

$$\text{irreversible work} \longrightarrow \text{persistent information}. \tag{62}$$

Let $\mathcal{P}_{\rm diss}(t)$ denote the total exergy dissipation rate of a localized, fault-tolerant information network. The cumulative dissipated physical energy is

$$E_{\rm diss}(t) = \int_{t_0}^{t} \mathcal{P}_{\rm diss}(t')\, dt'. \tag{63}$$

The accumulated work securing the ledger can be represented by a monotonic quantity $W_{\rm cum}(t)$, proportional to the integrated difficulty-weighted computational effort. A dimensionless measure of persistent information order is then

$$\mathcal{I}_{\rm PoW}(t) = \log\left(1 + \frac{W_{\rm cum}(t)}{W_0}\right), \tag{64}$$

where W_0 is a reference scale.

The logarithmic form reflects the fact that security is governed by relative cost: each additional unit of accumulated work increases the difficulty of rewriting the ledger by a multiplicative factor.

The rate of order accumulation is

$$\dot{\mathcal{I}}_{\rm PoW} = \frac{\dot{W}_{\rm cum}}{W_0 + W_{\rm cum}}. \tag{65}$$

This construction highlights a key feature: the system generates an effective arrow of time. Past states become increasingly resistant to modification as cumulative work grows.

An effective thermodynamic cost of persistent memory may be defined as

$$\mathcal{C}_{\rm mem} = \frac{E_{\rm diss}}{\mathcal{N}_{\rm state}}, \tag{66}$$

where $\mathcal{N}_{\rm state}$ denotes the number of cryptographically sealed macroscopic state transitions within the topological register.

The analogy with the orthogonal-value construction introduced earlier can be expressed through a localized functional:

$$\mathcal{V}_{\rm cons} = \left[\mathcal{E}_{\odot}^2 + \alpha_I \mathcal{I}_{\rm cons}^2\right]^{1/2}, \tag{67}$$

where $\mathcal{I}_{\rm cons}$ isolates the informational entropy reduction achieved through distributed consensus.

This expression mirrors the general vector norm structure

$$|\vec{V}| = \sqrt{a^2 + b^2}, \tag{68}$$

with thermodynamic embodiment and informational order contributing as orthogonal components. Fundamentally, physical energy does not strictly encode information in its own native units. Instead, we invoke a macroscopic analogue to Landauer's Principle [24]. While Landauer established the minimum thermal dissipation required to erase a bit, our bimetric interaction operates as the inverse generative process: the continuous, irreversible dissipation of exergy in $g_{\mu\nu}$ is the required physical cost to crystallize and seal topological information within $f_{\mu\nu}$ [28]. Recent phenomenological frameworks, such as the D-Trinity protocol combining cryptographic hashing (SHA-256) with decentralized relays, provide a topological blueprint for how such rigid informational ledgers achieve macroscopic thermodynamic stability and phase transitions [29].

In this sense, Proof-of-Work consensus mechanisms provide a localized laboratory-scale example of the general principle underlying the orthogonal informational sector: persistent informational order requires a physical substrate, an energetic cost, and an irreversibility mechanism. This analogy is strictly operational and does not imply any direct cosmological role.

A. Phenomenological Friction and the Bimetric Landauer Limit

In standard information theory, Landauer's principle defines the minimum thermal cost of erasing a bit [24]. We postulate an inverted, macroscopic analogue: the crystallization of topological order requires a continuous, irreversible expenditure of physical exergy.

Let $\delta W_{\rm diss}$ be the irreversible physical work dissipated in the visible metric $g_{\mu\nu}$, and $\delta\chi$ be the corresponding increment of negentropic order in $f_{\mu\nu}$. We hypothesize a fundamental threshold relation:

$$\delta W_{\rm diss} = \mathcal{T}^* \delta\chi, \tag{69}$$

where $\mathcal{T}^*$ acts as an effective topological tension (with dimensions of energy per unit order), representing the critical thermodynamic cost required to rigidly anchor an informational state against natural entropic decay.

In the language of field theory, the spontaneous evolution of the order parameter χ is governed by the Euler-Lagrange equation. To maintain the system away from thermal equilibrium (preventing the decay of χ down its potential U), the physical dissipation must act as an effective restorative force. We capture this by introducing a dissipative current $J_{\rm diss}$ into the scalar equation of motion:

$$\Box_f \chi - \frac{dU}{d\chi} + \frac{1}{\sqrt{-f}} \frac{\delta S_{\rm mix}}{\delta\chi} = J_{\rm diss}. \tag{70}$$

We model this current covariantly as being strictly sourced by the entropy production rate Φ_g within the physical metric. Taking n^λ as the unit 4-velocity vector of the macroscopic informational flow, the friction current is:

$$J_{\rm diss} = -\gamma(\Phi_g)\, n^\lambda \nabla_\lambda \chi, \tag{71}$$

where $\gamma(\Phi_g)$ is the bimetric friction coefficient, strictly monotonically increasing with the physical power dissipated in $g_{\mu\nu}$.

Crucially, in the rest frame of the homogeneous FLRW background of the f-metric ($ds_f^2 = -N_f^2 dt^2 + b^2 d\vec{x}^2$), the unit normal vector is $n^\lambda = (1/N_f, 0, 0, 0)$. Therefore, the directional derivative is exactly:

$$n^\lambda \nabla_\lambda \chi = \frac{1}{N_f}\dot{\chi} \equiv \chi'. \tag{72}$$

This mathematically proves that the friction current is coupled directly to the discrete topological time rate θ (the block clock), rather than the Newtonian time t. Substituting this along with the Hassan–Rosen interaction derivative yields the complete effective friction equation:

$$\Box_f \chi + \gamma(\Phi_g)\chi' - \frac{dU}{d\chi} + \frac{m_0^2}{M_{\rm Pl}}\beta \exp\left(-\frac{\beta\chi}{M_{\rm Pl}}\right) e_n\left(\sqrt{g^{-1}f}\right) = 0. \tag{73}$$

This equation dictates the macroscopic phase transition of the negentropic sector. If the physical dissipation Φ_g is below a critical threshold, the friction term $\gamma(\Phi_g)\chi'$ is insufficient to overcome the natural potential gradient $dU/d\chi$, leaving the informational state fluid and non-consensual. However, when Φ_g exceeds this threshold, the thermodynamic friction dominates. The field χ is dynamically frozen into a high-negentropy metastable state ($\chi' \to 0$), generating the rigid topological pressure required to source both the effective graviton mass and late-time cosmic acceleration.

X. MINIMAL EFFECTIVE COSMOLOGICAL SECTOR

To make the phenomenological content explicit, we introduce a minimal effective negentropic-sector action. The aim is not to derive a fundamental theory of the information metric $f_{\mu\nu}$ from microscopic quantum gravity, but to provide a controlled parameterization of its two leading cosmological effects: localized pressureless excitations and a slowly varying order condensate.

We write the informational action as:

$$S_{\mathcal{I}} = \int d^4x\sqrt{-f}\Bigg[-\frac{1}{2}f^{\mu\nu}\partial_\mu X \partial_\nu X - \frac{1}{2}m_X^2 X^2 - \frac{1}{2}f^{\mu\nu}\partial_\mu\chi\partial_\nu\chi - U(\chi)\Bigg], \tag{74}$$

where X denotes a massive informational excitation (topological inertia) and χ denotes the evolving information-order parameter.

For a homogeneous Friedmann background, utilizing the topological time derivatives ($X' \equiv \dot{X}/N_f$), the energy density and pressure of X evaluated on the f-manifold are:

$$\rho_X = \frac{1}{2}(X')^2 + \frac{1}{2}m_X^2 X^2, \tag{75}$$

$$p_X = \frac{1}{2}(X')^2 - \frac{1}{2}m_X^2 X^2. \tag{76}$$

If the topological mass is large ($m_X \gg H_f$), the field oscillates rapidly compared with the informational Hubble time. Averaging over oscillations gives:

$$\langle p_X \rangle \simeq 0, \qquad \langle \rho_X \rangle \propto b^{-3}. \tag{77}$$

Because the proportional scaling $b \propto a$ is enforced by the interaction potential, the projection into the visible sector yields $\rho_{\rm DM}^{\rm eff} \propto a^{-3}$. Thus, the massive excitation behaves precisely as cold dark matter at the physical background level.

Similarly, the macroscopic order field χ has:

$$\rho_\chi = \frac{1}{2}(\chi')^2 + U(\chi), \tag{78}$$

$$p_\chi = \frac{1}{2}(\chi')^2 - U(\chi). \tag{79}$$

Its effective equation-of-state parameter is therefore:

$$w_\chi = \frac{\frac{1}{2}(\chi')^2 - U(\chi)}{\frac{1}{2}(\chi')^2 + U(\chi)}. \tag{80}$$

In the friction-dominated metastable regime, driven by continuous physical exergy dissipation as derived in the previous section, the kinetic term is overdamped:

$$(\chi')^2 \ll U(\chi), \tag{81}$$

and one naturally obtains:

$$w_\chi \simeq -1, \tag{82}$$

so that the macroscopic order sector behaves as an effective dark-energy component.

The visible-sector Friedmann equation may then be written as:

$$3M_g^2 H_g^2 = \rho_{\rm SM} + \rho_X^{\rm eff} + \rho_\chi^{\rm eff} + \rho_{\rm int}, \tag{83}$$

where H_g is the visible Hubble rate and $\rho_{\rm int}$ encodes the bimetric mixing terms between $g_{\mu\nu}$ and $f_{\mu\nu}$ (including the χ-modulated effective cosmological constant). At the level of background cosmology, the model therefore reproduces the standard ΛCDM decomposition:

$$\rho_{\rm tot} = \rho_{\rm baryon} + \rho_{\rm radiation} + \rho_{\rm DM} + \rho_{\rm DE}, \tag{84}$$

with the rigorous geometric identification:

$$\rho_{\rm DM} \equiv \rho_X^{\rm eff}, \qquad \rho_{\rm DE} \equiv \rho_\chi^{\rm eff} + \rho_{\rm mix}^\Lambda. \tag{85}$$

This construction is deliberately conservative. It does not require the cosmological informational sector to be directly identified with specific terrestrial digital systems. Instead, the fields X and χ are effective degrees of freedom whose phenomenology—grounded in the thermodynamics of irreversible state transitions—can be formally constrained by standard cosmological observations.

XI. OBSERVATIONAL CONSTRAINTS AND FALSIFIABILITY

The model is intended as an effective parameterization and must therefore be constrained by the same observations that constrain ordinary dark matter, quintessence, and bimetric gravity. At minimum, a viable region of parameter space must satisfy the following conditions.

First, the effective dark-matter component must be sufficiently cold:

$$c_{s,X}^2 \ll 1 \tag{86}$$

during the epoch of structure formation.

Second, the order-field equation of state must remain close to that of a cosmological constant at late times:

$$w_\chi(z=0) = -1 + \epsilon, \qquad |\epsilon| \ll 1. \tag{87}$$

Third, the dynamic bimetric mixing dictates unique signatures in gravitational-wave (GW) astrophysics. The interaction between $g_{\mu\nu}$ and $f_{\mu\nu}$ induces graviton oscillations over cosmological distances. This implies a testable discrepancy between the GW luminosity distance ($d_L^{\rm GW}$) and the electromagnetic luminosity distance ($d_L^{\rm EM}$):

$$\frac{d_L^{\rm GW}(z)}{d_L^{\rm EM}(z)} \neq 1. \tag{88}$$

Current constraints from LIGO/Virgo multi-messenger events (e.g., GW170817) and future LISA observations place stringent bounds on the bare mass m_0 and the coupling β. Furthermore, the propagation speed of tensor perturbations must strictly satisfy $|(c_T^2/c^2) - 1| \lesssim 10^{-15}$.

Fourth, the effective Newtonian potential must reproduce standard large-scale structure (LSS) formation, governed by:

$$\nabla^2\Phi = 4\pi G_{\rm eff}(\chi, a)\left(\rho_{\rm baryon} + \rho_X^{\rm eff}\right). \tag{89}$$

Because $G_{\rm eff}$ is dynamically modulated by the informational order field χ, the growth rate of structure, parameterized by $f\sigma_8$, will exhibit characteristic redshift-dependent deviations from the ΛCDM prediction.

Additionally, models with two independent metrics naturally source isocurvature perturbations. Data from the Planck satellite restricts the amplitude of Cold Dark Matter isocurvature modes to be tightly sub-dominant to adiabatic fluctuations, $\mathcal{P}_{\rm iso}/\mathcal{P}_{\rm ad} \ll 1$. The initialization of the $f_{\mu\nu}$ metric post-inflation must respect these cosmic microwave background (CMB) bounds.

These requirements make the model falsifiable. It is disfavored if the informational excitation suppresses structure formation, if the order field deviates strongly from $w \simeq -1$, or if the sector coupling induces observable deviations from general relativity.

Parameter	Meaning	Phenomenological role
m_X	excitation mass	controls cold-DM behaviour
$U(\chi)$	order-field potential	controls dark-energy behaviour
$\mathcal{T}^*$	topological tension	threshold for thermodynamic friction
α_I	information weight	weights orthogonal order
W_0	reference work scale	normalizes macroscopic consensus
β	scalar coupling	controls dynamic bimetric mass $m^2(\chi)$
M_f/M_g	Planck mass ratio	sets informational gravity scale

TABLE I. Minimal parameter set of the effective orthogonal-bimetric model.

XII. LINEAR PERTURBATIONS AND STABILITY CONDITIONS

A viable model must be stable at the perturbative level. We write scalar perturbations of the visible metric in Newtonian gauge:

$$ds_g^2 = -(1+2\Phi)dt^2 + a^2(t)(1-2\Psi)d\vec{x}^{\,2}. \tag{90}$$

For the informational $f_{\mu\nu}$ metric, incorporating the topological lapse function N_f:

$$ds_f^2 = -N_f^2(1+2\Phi_f)dt^2 + b^2(t)(1-2\Psi_f)d\vec{x}^{\,2}. \tag{91}$$

The scalar perturbations of the informational degrees of freedom are:

$$\chi(t,\vec{x}) = \bar{\chi}(t) + \delta\chi(t,\vec{x}), \tag{92}$$

$$X(t,\vec{x}) = \bar{X}(t) + \delta X(t,\vec{x}). \tag{93}$$

The quadratic action for perturbations takes the schematic form:

$$S^{(2)} = \frac{1}{2}\int dt\, d^3k\, \left[\dot{\vec{q}}^T K \dot{\vec{q}} - \vec{q}^T \Omega^2 \vec{q}\right], \tag{94}$$

where the perturbation vector is

$$\vec{q} = (\delta\chi, \delta X, \Phi, \Psi, \Phi_f, \Psi_f). \tag{95}$$

Ghost freedom (absence of the Boulware–Deser ghost) requires the kinetic matrix to be positive definite, $K > 0$. Gradient stability requires strictly positive sound speeds, $c_{s,i}^2 > 0$.

Tachyonic instabilities are avoided when the relevant eigenvalues of the effective mass matrix satisfy:

$$m_{{\rm eff},i}^2 \gtrsim -H_g^2. \tag{96}$$

Crucially, in standard bimetric theories, the scalar sector is notoriously susceptible to the Higuchi ghost or tachyonic

collapse. In our thermodynamic framework, the massive physical dissipation (Φ_g) that freezes the background field $\bar{\chi}$ also exerts a profound fluctuation-dissipation drag on the perturbations $\delta\chi$. This continuous topological friction provides a natural stabilizing mechanism against catastrophic instabilities in the informational scalar sector.

The tensor sector must also be strictly stable. For transverse traceless tensor modes h^g_{ij} and h^f_{ij}, the quadratic action rigorously separates the proper volume elements and time derivatives (recalling $h'_f = \dot{h}_f/N_f$):

$$S_T^{(2)} = \frac{1}{8}\int dt\, d^3k \left[a^3 M_g^2 \left(\dot{h}_g^2 - c_T^2 \frac{k^2}{a^2} h_g^2 \right) + N_f b^3 M_f^2 \left((h'_f)^2 - c_{T,f}^2 \frac{k^2}{b^2} h_f^2 \right) - a^3 m_T^2(\bar{\chi}) (h_g - h_f)^2 \right]. \tag{97}$$

Compatibility with multi-messenger gravitational-wave observations requires the physical tensor propagation speed to be strictly luminal, $c_T \approx 1$, and places severe bounds on the effective mass $m_T(\bar{\chi})$.

A. Scalar Perturbations and Growth of Structure

To assess phenomenological viability, we focus on sub-horizon scales, where $k \gg aH_g$, and assume that the negentropic sector couples to visible matter strictly gravitationally (negligible portal couplings). In this regime, the dominant effect of the orthogonal sector is captured through an effective modification of the physical Poisson equation:

$$k^2\Phi = -4\pi G_{\rm eff}(k,a) a^2 \rho_m \delta_m. \tag{98}$$

The effective gravitational coupling is parameterized as:

$$G_{\rm eff}(k,a) = G\left[1 + \mu(k,a)\right], \tag{99}$$

where $\mu(k,a)$ encodes the fifth-force contribution mediated by the bimetric scalar degree of freedom.

For viable parameter ranges, we require:

$$|\mu(k,a)| \ll 1 \tag{100}$$

at late times, ensuring consistency with large-scale structure (LSS) and weak lensing observations.

The physical matter density contrast obeys:

$$\ddot{\delta}_m + 2H_g\dot{\delta}_m - 4\pi G_{\rm eff}\rho_m\delta_m = 0. \tag{101}$$

Thus, the orthogonal sector modifies structure growth primarily through $G_{\rm eff}$. A small positive μ enhances clustering, while a negative value suppresses it.

The scalar order field χ contributes primarily at the background level through its effective equation of state:

$$w_\chi(z) = -1 + \delta w(z), \tag{102}$$

with $|\delta w(z)| \lesssim \mathcal{O}(10^{-2})$ enforced by the thermodynamic friction, ensuring consistency with current cosmic microwave background (CMB) and supernovae constraints.

Matching the observed dark-energy density implies:

$$U(\chi_0) \sim (10^{-3}\,{\rm eV})^4, \tag{103}$$

which sets the characteristic scale of the informational order-field potential.

A full treatment requires solving the coupled perturbation system for $(\delta\chi, \delta X, \Phi, \Psi, \Phi_f, \Psi_f)$. However, the effective parameterization above captures the leading observable effects: a small, testable modification of gravitational clustering ($f\sigma_8$) and a friction-stabilized, nearly cosmological-constant background.

XIII. MACROSCOPIC INTERPRETATION

The cosmological phenomenology of this model can be distilled into the correspondence between an effective macroscopic state vector and the fundamental orthogonal bimetric tensor:

$$\vec{V} = \begin{pmatrix} a \\ b \end{pmatrix} \quad \longleftrightarrow \quad \mathcal{G}_{MN} = \begin{pmatrix} g_{\mu\nu} & 0 \\ 0 & f_{\mu\nu} \end{pmatrix}. \tag{104}$$

The a component, corresponding to the $g_{\mu\nu}$ metric axis, represents the geometry of thermodynamic embodiment: matter, radiation, chemistry, biology, and emergy-bearing physical structures. The b component, corresponding to the $f_{\mu\nu}$ metric axis, represents an orthogonal informational geometry: topological order, persistent memory, and non-electromagnetic data structures.

Within this framework, the dark sector is conceptually demystified as the macroscopic gravitational projection of the informational manifold. Specifically:

$$X \implies \text{Cold Dark Matter (Localized topological inertia resisting entropic erasure)}, \tag{105}$$

$$\chi \implies \text{Dark Energy (Global informational vacuum pressure frozen by thermodynamic friction)}. \tag{106}$$

The vector norm, representing the total magnitude of the coupled system,

$$|\vec{V}| = \sqrt{a^2 + b^2}, \tag{107}$$

is expressed macroscopically through the Bimetric Value Functional:

$$\mathcal{V}_{\rm HB} = \int_{\Sigma_t} d^3x \sqrt{\gamma_g} \left[\mathcal{E}_\odot^2 + \alpha_I \mathcal{I}^2 \right]^{1/2}. \tag{108}$$

This expression captures the central philosophical and physical claim of the theory: reality is not exhausted by the material axis alone (reductionist materialism), nor by information alone (the pure simulation hypothesis), but by their dynamically normed synthesis. The universe clusters and accelerates because it is actively preserving its own structural and topological history at a continuous thermodynamic cost.

XIV. QUANTUM DECOHERENCE AS MICRO-BIMETRIC INTERACTION

While the macroscopic dynamics of the orthogonal-bimetric framework explain dark sector phenomenology, the bimetric structure intrinsically accommodates the conceptual foundations of quantum mechanics. In standard quantum theory, the state of a system prior to measurement is described by a complex probability amplitude, $\psi(x,t) = \psi_R + i\psi_I$. Within our geometric correspondence, where the state tensor separates into orthogonal physical and informational metrics ($\mathcal{G}_{MN} = g_{\mu\nu} \oplus f_{\mu\nu}$), this complex nature is no longer a purely statistical artifact, but a reflection of the underlying bimetric topology.

We postulate that quantum superposition represents a regime where an excitation propagates predominantly within the informational metric $f_{\mu\nu}$, unconstrained by the macroscopic thermodynamic dissipation of the physical axis. In this state, the particle explores the non-local topological graph of the informational sector, providing a geometric basis for quantum entanglement. Particles separated by vast spatial distances in $g_{\mu\nu}$ may remain topologically adjacent in $f_{\mu\nu}$.

The act of quantum measurement (decoherence) requires a macroscopic apparatus, inevitably involving irreversible thermodynamic dissipation in the visible metric. Thus, wavefunction collapse is interpreted as a micro-scale phase transition induced by the bimetric mixing potential S_{mix}. The thermal friction of the measurement forces the delocalized informational state in $f_{\mu\nu}$ to precipitate onto the physical axis $g_{\mu\nu}$. Decoherence is, therefore, the microscopic inverse of cryptographic consensus: thermal dissipation forcing a projection from the informational axis ($f_{\mu\nu}$) down to the physical axis ($g_{\mu\nu}$). This geometric interpretation aligns with relational quantum mechanics and provides a unified origin for both quantum indeterminacy and cosmological dark matter.

XV. LIMITATIONS

The present model is speculative and effective. It does not derive the metric $f_{\mu\nu}$ from a fundamental microscopic theory of quantum gravity. It also does not prove that information is literally geometric. Rather, it provides a phenomenological parameterization in which informational order is represented gravitationally.

Several limitations must be emphasized.

First, the block-diagonal tensor notation is formal. Physical observables remain projections onto the real thermodynamic subspace.

Second, the emergy and bimetric value functional are macroscopic interpretive constructs, rather than fundamental sources explicitly inserted into the Einstein field equations.

Third, the extension of this macroscopic bimetric framework to quantum decoherence remains phenomenological. The model does not yet provide a modified non-linear Schrödinger equation, nor a specific Lindblad master equation, to mathematically formalize how the bimetric mixing potential S_{mix} induces objective state reduction. Deriving the micro-scale energy dissipation of wave-function collapse strictly from the $f_{\mu\nu}$ metric requires a comprehensive quantum-gravitational treatment.

Fourth, localized cryptographic consensus architectures serve merely as rigorous thermodynamic analogies for the irreversible dissipation required to secure informational topology. They are not theorized to source cosmological dark matter.

Finally, a full non-linear perturbative analysis is required. The model must ultimately be confronted quantitatively with CMB isocurvature bounds, baryon acoustic oscillations, and gravitational-wave phase deviations.

XVI. CONCLUSION

We have presented an effective orthogonal-bimetric model in which the physical metric $g_{\mu\nu}$ is supplemented by an additional metric $f_{\mu\nu}$ describing an orthogonal negentropic sector. The resulting framework provides a minimal phenomenological description in which localized informational inertia (massive excitations) reproduces cold dark matter behavior, while a macroscopic order parameter, dynamically frozen by thermodynamic friction, generates accelerated cosmic expansion.

The block-diagonal tensor representation

$$\mathcal{G}_{MN} = g_{\mu\nu} \oplus f_{\mu\nu} \tag{109}$$

serves as a formal organizational tool to enforce orthogonality between matter and information, preventing trivial complex superpositions rather than acting as a literal 8D extension of macroscopic spacetime. Physical observables remain firmly associated with the physical sector.

The model is constructed to be testable. Viability requires consistency with structure formation, cosmic expansion history, gravitational-wave constraints, and the absence of observable fifth forces. These conditions restrict the allowed parameter space of the negentropic sector.

An interpretive layer was introduced in which the $f_{\mu\nu}$ manifold is associated with maintained informational order. Distributed cryptographic consensus (Proof-of-Work) was utilized as a rigorous, localized thermodynamic analogue, demonstrating how irreversible physical dissipation acts as a geometric friction that crystallizes and secures

topological data. While strictly illustrative at the cosmological scale, it grounds the model's fundamental premise: topological order requires continuous energetic expenditure.

Several open problems remain. A full perturbative analysis is required to establish stability and determine the detailed evolution of density perturbations. In addition, a deeper connection to fundamental theories of quantum gravity or statistical mechanics is needed to derive the information sector from first principles.

Despite these limitations, the framework provides a controlled, falsifiable phenomenological setting in which dark matter, dark energy, and information-theoretic thermodynamics can be synthesized within a unified effective description.

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